PROJECT PORTFOLIO · REMOTE SENSING

Iceberg Alley SAR

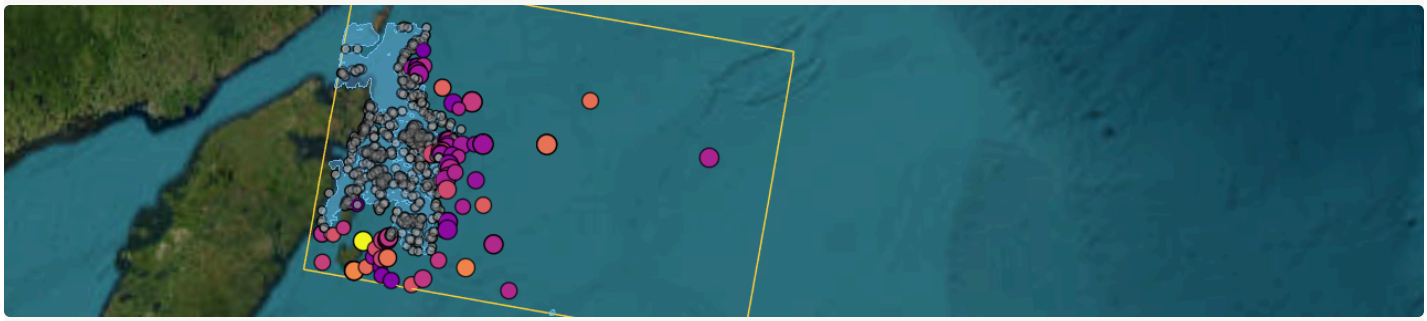
Detecting icebergs off Newfoundland in Sentinel-1 satellite radar with a classical detector and a foundation model, validated against International Ice Patrol data and shown on a live 3D globe.

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Independent project, 2026

Code: `github.com/natali-kovalski/iceberg-alley-sar`

Live demo: `natali-kovalski.github.io/iceberg-alley-sar`



The viewer, 2 May 2025: scene footprint (yellow), pack ice (blue), candidates coloured by radar contrast, likely ice floes in grey.

78%
CA-CFAR recall against Ice Patrol analyst labels on the same pass (207 of 267)

67 m
median position difference to the labelled bergs

+10 pts
recall for the THOR foundation model over the CFAR, at the same detection count

32
Sentinel-1 passes (2019–2026) in the live viewer, CFAR and THOR layers

The problem

Each spring, icebergs from Greenland drift down the Labrador Current past northeast Newfoundland, into shipping lanes, fishing grounds and offshore platform areas. Aircraft patrols are costly and grounded by fog. Synthetic aperture radar (SAR) images the ocean through cloud and darkness. The question: how reliably can free Sentinel-1 data find these bergs?

Approach

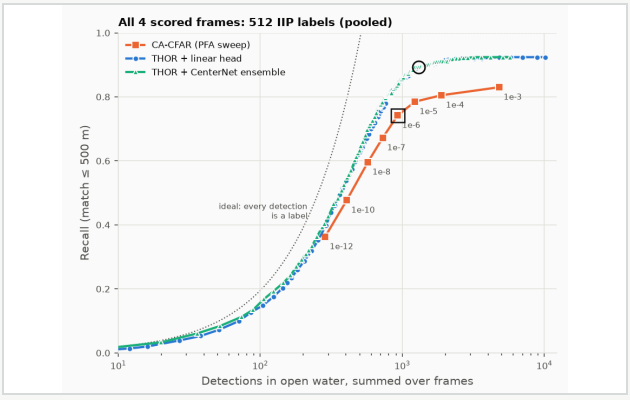
1. **Search and order.** HH/HV scenes from ASF, calibrated and terrain-corrected by HyP3 RTC at 10 m.
2. **Mask.** Land from OpenStreetMap plus a 500 m buffer; pack ice from cross-pol (HV) texture on 200 m cells.
3. **Detect.** Cell-averaging CFAR on HV, threshold from a 10^{-6} false-alarm rate and the scene's speckle.
4. **Filter.** Size limits, rejection of thin sea-ice strips, and an HH co-polarization check that removes speckle at 10 m.
5. **Publish.** GeoJSON with per-target attributes, then CZML for a CesiumJS viewer on GitHub Pages.

Validation

Ground truth is scarce, so I tested four sources: ice charts, aircraft sightings, Sentinel-2 optical imagery, and Ice Patrol analyst labels on the same radar pass. Only the last supports a per-berg score (65% recall for small, 81% for medium bergs); the code reports the others as unresolved instead of forcing a score.

A foundation model as a second detector

I ran **THOR**, a SAR foundation model, frozen, with a 769-parameter linear head trained on Ice Patrol labels from 2019–2020. On four held-out frames (512 labelled bergs) it found **84%** where the CFAR found **74%** at the same number of detections, and 11 of 14 large bergs where the CFAR's strip filter kept 4. It still needs the CFAR pipeline's pack-ice mask.



Recall vs. detections on 4 held-out April 2019 frames: THOR heads (blue, green) vs. the CA-CFAR swept over its false-alarm rate (orange).

Engineering practice

- **Modular Python package** with type hints, a CLI and 153 tests
- **One config file**, lengths in metres so tuning holds across resolutions
- **Re-runnable pipeline** that shows processing cost first
- **Live viewer** rebuilt by one command, deployed by CI

Limits and next steps

Precision needs AIS ship positions to measure. Next: AIS matching, a shape-based strip filter so the CFAR keeps large bergs, and THOR at 4×4 patches for small bergs.

TOOLS

`Python · NumPy · SciPy · rasterio · GeoPandas · PyTorch · TerraTorch · HyP3 · Vue 3 · CesiumJS`

Data: modified Copernicus Sentinel data (2019–2026) · International Ice Patrol (NSIDC G00807) · North American Ice Service · THOR: FM4CS (Apache 2.0) · © OpenStreetMap contributors